
TENSOR NETWORK REPRESENTATION OF FOURIER TRANSFORMS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

September 29, 2026

ABSTRACT

The Fourier transform is represented by a tensor, when unfolding the vector transform using a binary coordinate representation. This tensor admits a sparse decomposition into (controlled) unitary matrices. We derive two standard algorithms in the contraction formalism:

- Fast Fourier Transform, by layerwise contraction of the tensor network
- Quantum Fourier Transform, by interpreting the tensor network as a quantum circuit

1 Unfolding

The Fourier transform of vectors in $\mathbb{C}^m$ is represented by a matrix

$$F[X = x, Y = y] = \frac{1}{\sqrt{m}} \exp \left[2\pi i \cdot \frac{x \cdot y}{m} \right]$$

where X, Y are variables of dimension m . The Fourier transform of a vector $V[X]$ is then the contraction

$$\langle V[X], F[X, Y] \rangle_{[Y]} .$$

We assume $m = 2^d$ and represent each index $x \in [2^d]$ by a tuple $(x_0, \dots, x_{d-1}) \in [2]^d$ such that

$$x = \sum_{k \in [d]} x_k \cdot 2^{d-k-1} .$$

Note that in this convention, x_{d-1} is the least significant bit. We follow Nielsen (2010) and denote x by $x_0 \dots x_{d-1}$. By $x_0 \dots x_{k-1} . x_k \dots x_{d-1}$ we denote $\frac{1}{2^{d-k-1}} \cdot x$.

The unfolded vector $V[X]$ is a tensor $V[X_{[d]}]$ with indices

$$V[X_{[d]} = x_{[d]}] = V[X] = \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1} \right) .$$

In the unfolded representation the fourier transform is described as the contraction with a tensor $F^d[X_{[d]}, Y_{[d]}]$ with coordinates

$$F^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \frac{1}{\sqrt{2^d}} \exp \left[2\pi i \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1} \right) \cdot \left(\sum_{k \in [d]} y_k \cdot 2^{d-k-1} \right) \right] ,$$

that is the Fourier transform of $V[X_{[d]}]$ is

$$\langle V[X_{[d]}], F[X_{[d]}, Y_{[d]}] \rangle_{[Y_{[d]}]} .$$

2 Tensor network decomposition

We now derive a tensor network decomposition of the tensor $F^d[X_{[d]}, Y_{[d]}]$.

Lemma 1. *The Fourier transform of basis vectors $\epsilon_{x_{[d]}}[X_{[d]}]$ is the elementary tensor*

$$\frac{1}{\sqrt{2^d}} \bigotimes_{k \in [d]} (\epsilon_0[Y_k] + \exp[2\pi i \cdot (0.x_{d-k-1} \cdots x_{d-1})] \cdot \epsilon_1[Y_k])$$

Proof. For each $y_{[d]}$ we have

$$\begin{aligned} & \langle \{(\epsilon_0[Y_k] + \exp[2\pi i \cdot 0.x_{d-k-1} \cdots x_{d-1}] \cdot \epsilon_1[Y_k]) : k \in [d]\} \cup \{\epsilon_{y_{[d]}}[Y_{[d]}}]\} \rangle_{[\emptyset]} \\ &= \prod_{k \in [d]} \exp[2\pi i \cdot (0.x_{d-k-1} \cdots x_{d-1}) \cdot y_k] \\ &= \prod_{k \in [d]} \exp[2\pi i \cdot (x_0 \cdots x_{d-k-2} \cdot x_{d-k-1} \cdots x_{d-1}) \cdot y_k] \\ &= \exp \left[2\pi i \cdot x \sum_{k \in [d]} y_k \cdot 2^{d-k-1} \right] \\ &= \exp \left[2\pi i \cdot \frac{x \cdot y}{2^d} \right]. \end{aligned}$$

Since the basis coefficients of the elementary tensor coincide with those of the Fourier transform of $\epsilon_{x_{[d]}}[X_{[d]}]$, both tensors are equal. $\square$

We introduce the diagonal unitary

$$U^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \begin{cases} \exp \left[2\pi i \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{-k-2} \right) \right] & \text{if } x_{[d]} = y_{[d]} \\ 0 & \text{else} \end{cases}$$

Note that U^d is diagonal and the tensor product of matrices with variables X_k, Y_k . Diagonality can be used to contract U^d to an arbitrary tensor with computational demand in $\iota(2^d)$.

Using diagonality and elementarity, we have

Lemma 2. *The Fourier transform on $[2^d]$ can be decomposed into a Fourier transform on $[2^{(d-1)}]$ as:*

Proof. It suffices to show that the Fourier transform of arbitrary basis tensors $\epsilon_{x[d]} [X_{[d]}]$ follows the decomposition. The Fourier transform of $\epsilon_{[d-1]} [X_{[d-1]}]$ is by the Lemma above

$$\frac{1}{\sqrt{2^{d-1}}} \bigotimes_{k \in [d-1]} \left(\epsilon_0 [\tilde{Y}_k] + \exp [2\pi i \cdot (0.x_{d-k-2} \cdots x_{d-2})] \cdot \epsilon_1 [\tilde{Y}_k] \right)$$

In case of $x_{d-1} = 0$, the Fourier transform of $\epsilon_{x[d]} [X_{[d]}]$ is the tensor product of this state with $\epsilon^+ [X_0]$. This is reproduced by the decomposition, since the controlled unitary U^{d-1} acts trivially and the Hadamard gate transforms $\epsilon_0 [d-1]$ to $\epsilon^+ [X_0]$.

In case of $x_{d-1} = 1$, the leg vectors $\left(\epsilon_0 [\tilde{Y}_k] + \exp [2\pi i \cdot (0.x_{d-k-2} \cdots x_{d-2})] \cdot \epsilon_1 [\tilde{Y}_k] \right)$ are further transformed into

$$(\epsilon_0 [Y_{k+1}] + \exp [2\pi i \cdot (0.x_{d-k-2} \cdots x_{d-2}x_{d-1})] \cdot \epsilon_1 [Y_{k+1}]) .$$

In both cases we arrive at the Fourier transform of $\epsilon_{x[d]} [X_{[d]}]$. $\square$

By iteratively decomposing F^{d-1} until we arrive at F^1 (which is the H matrix), we get a decomposition of F^d into d H matrices and controlled U unitaries.

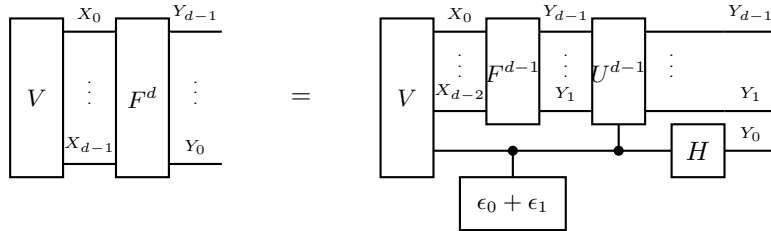
3 Fast Fourier Transform

3.1 Layerwise contraction

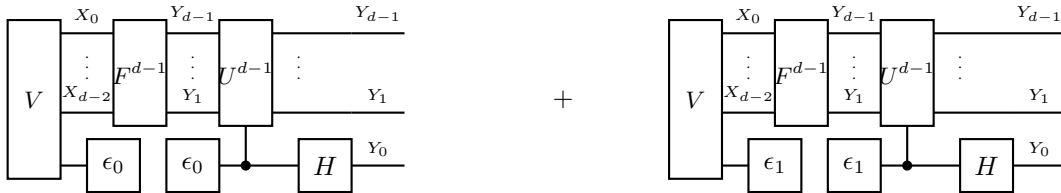
We contract the FFT tensor network layerwise. There are $2 \cdot d$ layers, each has a contraction demand in $\mathcal{O}(2^d)$ (exploiting diagonality of U). The algorithm has a total computational demand in $\mathcal{O}(d2^d)$.

3.2 Divide and conquer

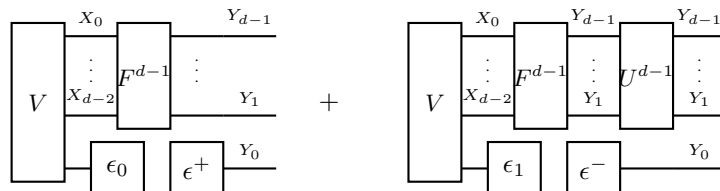
Called the radix-2 FFT Cooley and Tukey (1965). Can be seen as an iterative simplification by adding a $\mathbb{I} [X_{d-1}]$ and splitting it into the basis vectors.



Now by linearity of contractions, this splits into two terms



We simplify them to



The contraction of V with $\epsilon_1[X_{d-1}]$ respectively $\epsilon_1[X_{d-1}]$ are the even and the odd coordinates of V . Both are Fourier transformed with $d-1$ bits, which are further divided into Fourier transforms of lower order. In the conquer step of the FFT, the odd Fourier transform gets a coordinatewise transform by U , which is efficient since U is diagonal.

4 Quantum Fourier Transform

When $V[X_{[d]}]$ is a quantum state, the Fourier transform can be understood as a unitary evolving the state. Since both the controlled U tensor and the Hadamard H are unitary, each of them can be implemented by a sequence of gates. For sparse implementation one can exploit that U is a tensor product of matrices (i.e. single qubit gates).

To realize the QFT we need d single-qubit Hadamard gates and $\sum_{k \in [d]} k = \frac{d \cdot (d-1)}{2}$ controlled diagonal single-qubit gates (to realize the U^k transforms). The gate complexity is therefore $\mathcal{O}(d^2)$.

5 Generic groups

So far we have treated the Fourier transform on the cyclic group $[\prod_{k \in [d]} m_k]$ using unfoldings. More naturally, a tensor is defined as a complex field on the cartesian product group $\times_{k \in [d]} [m_k]$. Let us now generalize to generic groups.

5.1 Tensors as group fields

A group field is a vector

$$\tau[X] : G \rightarrow \mathbb{C}$$

where X enumerated the $|G| = m$ elements of the group. The x -th element of the group is then mapped to $\tau[X = x]$.

Example 1. *Unfolding is not an isomorphism* Let us note that the unfolding of $[\prod_{k \in [d]} m_k]$ to $\times_{k \in [d]} [m_k]$ is not a group isomorphism. The group operation in $\prod_{k \in [d]} m_k$ is the integer summation

$$x \circ y = x + y \mod \left(\prod_{k \in [d]} m_k \right).$$

In the unfolding the group operation is

$$(x_0, \dots, x_{d-1}) \circ (y_0, \dots, y_{d-1}) = (x_0 + y_0 \mod m_0, \dots, x_{d-1} + y_{d-1} \mod m_{d-1}).$$

Intuitively, the difference of these groups is that in the unfolded group summation carry bits are set to 0.

5.2 Representations

A representation is a map of G to a matrix group. Irreducible representations (irrep) ρ are those, which are not equivalent to matrix groups in block form. We represent them by tensors

$$\rho[X, Y^{\text{in}}, Y^{\text{out}}].$$

Example 2. *Cyclic group* For the cyclic group $G = [m]$ the irreps are one-dimensional, that is $n = 1$, and enumerated by $\ell \in [m]$ as

$$\rho^{[m_\ell]}[X = x, L = \ell, Y^{\text{in}} = 0, Y^{\text{out}} = 0] = \exp \left[2\pi i \cdot \frac{x \cdot \ell}{m} \right].$$

Note that for $m = 2$ this is the Hadamard matrix

$$\rho^{[2]}[X, L, Y^{\text{in}} = 0, Y^{\text{out}} = 0] = \sqrt{2} \cdot H[X, L].$$

5.3 Finite Abelian groups

Finite Abelian groups are equivalent to cartesian products of cyclic groups, that is

$$G \sim \bigtimes_{k \in [d]} [m_k] .$$

The irreps of finite Abelian groups are thus disconnected tensor networks

$$\rho^{\bigtimes_{k \in [d]} [m_k]} [X_{[d]}, L_{[d]}, Y_{\text{in}}^{[d]}, Y_{\text{out}}^{[d]}] = \bigotimes_{k \in [d]} \rho^{[m_k]} [X_k, L_k, Y_{\text{in}}^k, Y_{\text{out}}^k]$$

where each $\rho^{[m_k]} [X_k, L_k, Y_{\text{in}}^k, Y_{\text{out}}^k]$ is the irrep tensor to $[m_k]$, see Example 2.

5.4 Fourier transform

The Fourier transform of a group field $\tau [X]$ maps each irrep ρ to

$$\sqrt{\frac{n}{|G|}} \cdot \langle \tau [X], \rho [X, Y^{\text{in}}, Y^{\text{out}}] \rangle_{[Y^{\text{in}}, Y^{\text{out}}]} .$$

Example 3 (Walsh-Hadamard transform). *Consider the group $[2]^d$, and a complex field on it by a quantum state $\psi [X_{[d]}]$. The Fourier transform is then the application of Hadamard gates on each qubit, that is*

$$\left\langle \{ \psi [X_{[d]}] \} \cup \bigcup_{k \in [d]} \{ H [X_k, L_k] \} \right\rangle_{[L_{[d]}]} .$$

This is the Walsh-Hadamard transform of the quantum state.

6 Detecting periods

Based on the support of the Fourier transform, one can derive the periods of a function. Periods are indices $(Z_0, \dots, Z_{d-1})$ such that

$$\tau [X_{[d]} = x_{[d]}] = \tau [X_{[d]} = x_{[d]} \oplus z_{[d]}]$$

where $\oplus$ is the tuple position wise sum mod m_k .

Each supported index of the Fourier transform corresponds with periods in the function. The periods of the functions are thus at least the intersection of those (when assuming more structure one might have more periods).

6.1 Walsh-Hadamard transform

The Walsh-Hadamard transform is the Fourier transform with respect to the group $\bigtimes_{k \in [d]} [2]$. It is the legwise application of the Hadamard matrix.

In this case, we can easily determine the periods corresponding to each basis vector. The inverse Fourier transform of $\epsilon_{\ell_{[d]}} [L_{[d]}]$ is the tensor

$$\left(\bigotimes_{k \in [d] : \ell_k = 0} \epsilon^+ [X_k] \right) \otimes \left(\bigotimes_{k \in [d] : \ell_k = 1} \epsilon^- [X_k] \right) .$$

For each $\epsilon^+ [X_k]$ we have a period generator of ϵ_k . For each pair of $\epsilon^- [X_k] \otimes \epsilon^- [X_{\bar{k}}]$ we have a period generator of $\epsilon_k + \epsilon_{\bar{k}}$.

One example is the Deutsch-Jozsa test, where the transformed is the $\epsilon_0 [L_{[d]}]$ state, if and only if the function is constant. In this case, any vector is a period, since the group itself is generated by the period generators.

6.2 Application

QFT is a typical component in Quantum algorithms: QFT can be done efficiently with $\iota(d^2)$ controlled single-qubit gates. If the Fourier transform is a basis vector, a single measurement is sufficient to determine it.

References

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